

On semantic plurality within the Spanish DP and *ad sensum* agreement:
A Count-based mechanism for combinatorial variability*

Timothy Gupton (University of Georgia) & Chad Howe (University of Georgia)

Abstract

In Spanish *ad sensum* agreement, semantically-plural, but grammatically-singular subjects, of the form [DP1 [*de* NP2/DP2]] 'DP1 of NP2/DP2' (e.g. *la mayoría de los triunfadores* 'the majority of the winners'), agree with a singular or plural verb. Previous accounts identify a 'head N' responsible for agreement, but cannot explain variable agreement.

We couch our proposal within Distributed Morphology (DM), with number encoded and valued semantically and morphologically, following research on DP-internal gender (e.g. Kramer 2016, Puškar 2018). We propose that conflicting semantic and morphological number within the DP triggers *Count*, an alternative agreement mechanism that calculates a plural ratio diacritic for variable production based on feature values. We test our hypothesis comparing predicted rates of singular and plural verb agreement generated by *Count* with attested data in the *Corpus del Español* (CdE). Statistical analysis reveals that the variables in our proposal are significant predictors of variable agreement observed in the data.

(150 words)

Key words: variable number agreement, (pseudo)partitives, syntax, *ad sensum*, DP, Spanish

1. Introduction

In this article, we detail a proposal to account for *ad sensum* structures, which arise when a partitive (1) or pseudo-partitive (2) subject noun phrase in Spanish of the type [NP_1 [*de* NP_2]] 'NP₁ of NP₂' variably agrees with the verbal predicate. Consider these examples from the Royal Spanish Academy (RAE 2009) grammar.

- (1) a. *Un montón de fotógrafos de prensa registró el hecho.* (RAE 2009: 2573)

‘A heap (sg.) of press photographers (pl.) registered (sg.) the fact.’

- b. *Un montón de jugadores me metieron duro, no me regalaron nada...*

‘A heap (sg.) of players gave (pl.) it to me, they didn’t gift (pl.) me anything...’

- (2) a. *La mayoría de los triunfadores daba saltos.* (RAE 2009: 2578)

‘The majority (sg.) of the winners (pl.) jumped (sg.) around.’

- b. *La mayoría de los militares argentinos eran nacidos bajo el signo zodiacal Tauro.*

‘The majority (sg.) of the Argentine soldiers (pl.) were born (pl.) under the (zodiac) sign of Taurus.’

In (1a) the DP *un montón de fotógrafos de prensa* is the subject of the singular verb form *registró*, while in (1b), the DP *un montón de jugadores* is the subject of the plural verb form *metieron*.¹ In Spanish, the grammar is limited to the possibilities of singular or plural, but not all singular nouns are singular semantically (e.g. *grupo* ‘group’), nor are

all plural nouns plural semantically (e.g. *tijeras* ‘scissors’).² Although the RAE (2009) classifies the structures in (1) as pseudo-partitives, the examples in (2) show that partitive structures also participate in variable subject-verb agreement. These two examples share a structure with two nominal elements that differ in grammatical number: the higher, singular DP in *la mayoría* and a plural DP/NP modifier in *los triunfadores/los militares*.

Consider the DP *la mayoría de los militares*:

- (3) *la* *mayoría* *de los* *militares*
 DEF.F.SG majority.F.SG of DEF.M.PL soldiers.M.PL
 ‘the majority of the soldiers’

In (3), the structurally lower, yet linearly closer NP2 *militares* appears to agree with the verb in (2b), resulting in what is often referred to as *ad sensum* agreement (e.g. Brucart, 1997). In the following section, we review previous syntactic accounts of *ad sensum* phenomena, concluding that, despite their descriptive adequacy, previous syntactic proposals are lacking in explanatory power, in particular as it pertains to the predicted and attested production of such structures in linguistic corpora. In section 3, we present our model of variable agreement for *ad sensum* structures, which seeks to improve upon preceding proposals by adding a *Count* mechanism which tallies feature values when conflict arises between morphological and semantic number within the DP. The result of this mechanism is a diacritic that is added to the morphological specification of number on DP, thus informing the interfaces about the acceptable degree of morphological

variability in linguistic production. We propose that this is a rule-based phenomenon generated by the grammar, and not performance-related, therefore distinguishing it from *attraction*.³ The resulting diacritic, which is based on the presence of features within the syntax, generates testable predictions. In section 4, we test the predictions of the model with data from the *Corpus del Español* (Davies 2016). In section 5, we report on a statistical analysis of the corpus results, finding that linear regression analyses confirm that the factors predicted by *Count* to be operative in *ad sensum* agreement are, in fact, statistically significant variables predicting the generation of plural verb agreement. We offer concluding comments and directions for further research in the final section.

2. Previous accounts

The diachronic literature has dedicated a considerable amount of attention to *ad sensum* structures (e.g. Aarts 1998; Brems 2010; Traugott 2008a, 2008b), positing that constructions like English *a lot of NP* and *a sort of NP*, undergo a series of developments, beginning primarily as partitive constructions and then acquiring more general quantificational meanings. In the case of English structures like *a lot of*, these items are proposed to have grammaticalized into ‘binominal quantifiers’, collocating a range of NP₂ types (e.g., count and mass) and extending beyond the nominal domain to modify vPs, as in ‘I go to the library *a lot*’. In Verveckken (2012: 436) the Spanish *un montón de libros* ‘a heap of books’ is proposed to undergo a shift in literal meaning to a more abstract meaning equivalent to ‘a heap/a lot’. As part of this semantic change, she proposes that a structural reanalysis called ‘rebracketing’ takes place (4):

$$(4) [\text{Det} + \text{QN}_{\text{head}} + [\text{de} + \text{N2}] > [[\text{Det} + \text{QN} + \text{de}] + \text{N2}_{\text{head}}]$$

In Verveckken's proposal, QN refers to a 'quantifying noun' and the label 'head' designates the noun governing the entire DP. This proposal seeks to capture the semantic and syntactic change in the *ad sensum* construction, while also capturing the fact that the NP₂ may ostensibly agree with the verb (5).

- (5) *Un centenar de soldados*
 INDEF.M.SG hundred-some.M.SG of soldiers.M.PL
[detuvo / detuvieron] a los enemigos.
 detain.PST.3.SG / detain.PST.3.PL DOM DEF.M.PL enemy
 'a handful of soldiers stopped the enemy'

Although this sort of model attempts to explain how agreement may pass from the higher NP to the structurally lower NP, it does not explain why it sometimes does not happen in *ad sensum* structures.

Early work on partitives in the generative tradition (e.g. Jackendoff 1968, 1977) was more concerned with details related to the internal structure of the NP/DP than it was with subject-verb agreement. Previous accounts of *ad sensum* agreement phenomena have treated it as some flavor of “rebracketing”, such that the structurally lower NP₂

appears higher in the structure than NP1, thus justifying plural verb agreement instead of singular.

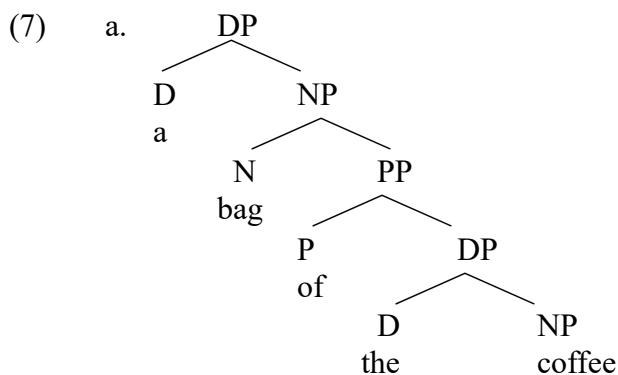
Selkirk (1977: 311) acknowledges the number ambiguity of subject-verb number agreement with examples like (6), but no mention of the locality problem enters the discussion.

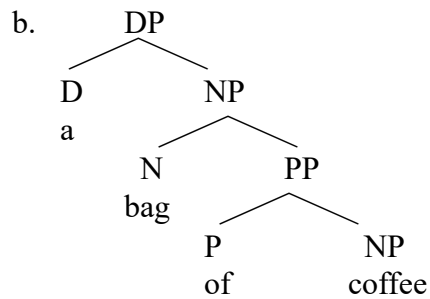
(6) Number agreement

A bunch of those flowers { was / were } thrown out on the back lawn.

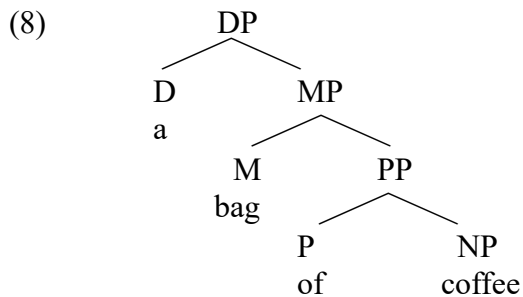
Stickney (2009) largely updates Selkirk's (1977) proposed X' syntactic structure.⁴

According to her proposal, the difference between partitives and pseudopartitives is determined by the internal modification structure, the former being characterized by a structurally lower DP modifier, and the latter by a lower NP modifier. N1-headed partitives, which exhibit agreement with the structurally higher N, are modified by DPs, and have the structure in (7a), while N1-headed pseudopartitives, which are modified by an NP, have the structure in (7b).





Cases in which the lower N2 head agrees with the verb are explained by a different structure in which the higher N1 is actually an MP (Measure Phrase), as we can see in (8).



Stickney (2009) adopts the functional projection Measure Phrase (MP, Alexiadou, Haegeman & Stavrou 2007) for English pseudo-partitives, proposing that the higher NP is an MP in a structural hierarchy ($DP_1 > MP > FP > DP_2$). This proposal is rather effective in explaining how a plural Number feature might appear to 'percolate' upward from the coda NP to the head NP—a preferable alternative to determining head NP via stipulation. Despite the fact that Stickney (2009) is supplemented by experimental data gathered from L1 English-acquirer children that attest the presence of variation, the deterministic model proposed does not consider how variation in agreement might result. Despite the fact that this model provides us with a fairly straightforward descriptive

solution for the locality problem as it relates to subject-verb agreement, it lacks predictive power regarding 1) when variation in subject-verb agreement of the type found in *ad sensum* structures should exist, and 2) to what degree we may expect variation to occur.

Brucart (1997) follows Selkirk (1977) in distinguishing partitive constructions from pseudo-partitives based on two key heuristics.⁵ The first is that partitives have a definiteness requirement within what Brucart calls the 'partitive coda', or the DP complement within the partitive structure, thus prohibiting (9a, cf. 9b-d).⁶

- (9) a. **Un grupo de* [_{DP} *unos* *manifestantes*]
a group of INDEF.M.PL protestors
'a group of some protestors'
- b. *Un grupo de* [_{DP} *cien* *manifestantes*]
a group of hundred protestors
'a group of a hundred protestors'
- c. *Un grupo de* [_{DP} *unos* *cien* *manifestantes*]
a group of INDEF.M.PL *hundred* protestors
'a group of approximately one hundred protestors'
- d. *Un grupo de* [_{DP} *unos* *pocos* *manifestantes*]
a group of INDEF.M.PL few protestors
'a group of a few protestors'

In this proposal, despite their similar structures, pseudo-partitives differ from partitives in that the coda DP (*niños* 'children' in (10)) must always be a bare nominal.

- (10) *Un grupo de niños*
 a group of children
 'a group of children'

Furthermore, true partitives allow extraposition of the partitive DP coda (11a, b), while pseudo-partitives do not (11c, d):

- (11) a. *De los manifestantes, un grupo ~~de los manifestantes~~ se dirigieron*
 of the protestors a group of the protestors SE address.PST.3.PL
al gobierno civil.
 DOM-ART.DEF.3.SG government civilian
 'of the protestors, a group addressed the civilian government'
- b. *De los manifestantes, el gobernador cree que la mayoría*
 of the protestors the governor think.PRS.3.SG that the majority
de los manifestantes eran estudiantes.
 of the protestors be.IMPFV.3.PL students
 'of the protestors, the Governor thinks that the majority were students'
- c. **De manifestantes, un centenar ~~de manifestantes~~ se dirigieron*
 of protestors a hundred-some of protestors SE address.PST.3.PL

al gobierno civil.

DOM-ART.DEF.3.SG government civilian

'of protestors, a hundred engaged with the civilian government'

- d. **De manifestantes, el gobernador cree quesólo un centenar*

of protestors the governor think.PRS.3.SG thatonly a hundred-some

~~*de manifestantes*~~ *eran estudiantes.*

of protestors be.IMPFV.3.PL students

'of the protestors the Governor thinks that only about a hundred or so were students'

This extraction asymmetry appears to be the prime motivator of Brucart's analysis of partitives and pseudo-partitives, leading to an account that moves partitive coda DPs to an operator position adjoined to the DP at Logical Form (LF). Given that this is a semantic issue, he reasons that it should be resolved via covert movement at LF.

Although this proposal solves the locality problem as it relates to verb agreement in partitive codas, this is only part of the story. Pérez-Jiménez & Demonte (2017) correctly note that partitives (12a=2a) as well as pseudo-partitives (12b) trigger *ad sensum* agreement.⁷

- (12) a. *La mayoría de los triunfadores daba saltos.*

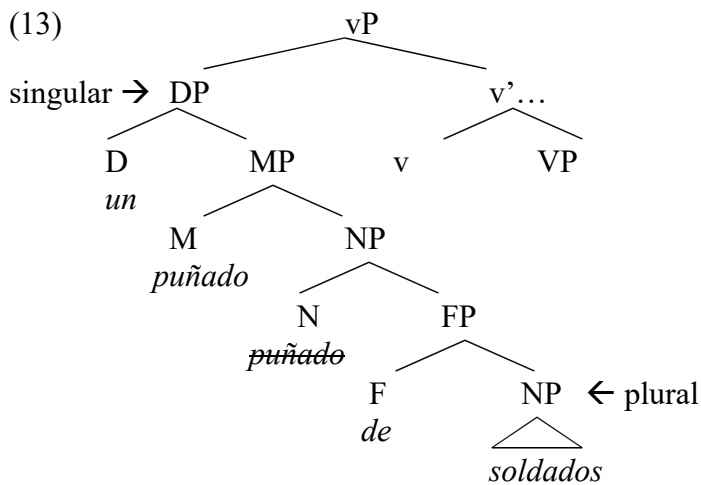
DEF.F.SG majority.F.SG of DEF.M.PL winners.M.PL give.IMPFV.3SG jumps

'The majority of the winners jumped around.'

- b. *Un puñado de soldados detuvo / detuvieron*
 DEF.M.SG handful.M.SG of soldiers.M.PL detain.PST.3SG / detain.PST.3.PL
a los enemigos.
 DOM DEF.M.PL enemy.M.PL
 'A handful of soldiers detained the enemy.'

In short, the fact that both structures trigger variable singular or plural agreement limits the explanatory power of Brucart's account. This is perhaps a welcome result, given the implications that it would hold for the proposed re-conception of the semantic module, which would find itself tasked with some subset of grammatical agreement calculations.

Let us briefly examine the conundrum arising from attested *ad sensum* constructions like *un puñado de soldados* from (12b) in (13).⁸



Assuming our simplified tree in (13), we see that the NP *un puñado* is hierarchically higher than the NP *soldados* within the partitive FP. Upon merger of the structure in (13)

with a finite T, the DP *un puñado de soldados* should already bear a value for number. Following Chomsky's (1998) Agree-based model, it is this number that the subject values on T when successfully probed by T. This is how subject-verb agreement is standardly understood to happen (although see e.g. Zwart, 2006 for an alternative proposal). Similarly, the ϕ -feature value for number on the determiner (D) may be understood to be supplied by the nominal NP selected by D, here singular *puñado*. Within this structural configuration it is impossible for the lower, plural DP *soldados* to agree with finite T, given that it is too deeply embedded in the hierarchical structure to be able to enter into a mutual c-command relationship with D and provide a plural value for D. Nevertheless, as we have seen, the empirical facts suggest otherwise. The challenge then, is to find an alternate model of the syntax of partitive DPs that can 1) provide an account for why and how *ad sensum* agreement occurs, but also 2) not generate any of the impossible instances of agreement mismatch. This is no small undertaking. We find it an undesirable prospect to dismiss occurrences of mismatched agreement in *ad sensum* structures as performance errors; rather, we feel that it is preferable to consider improvements to the syntactic model that can allow it to account for such variation. The following two models of the syntax attempt to do just that.

The first of these is Pérez-Jiménez & Demonte (2015, 2017), which assumes a feature-sharing approach similar in spirit to Frampton & Gutmann (2006).⁹ Although this proposal focuses on Spanish, Pérez-Jiménez & Demonte note that *ad sensum* agreement phenomena are also attested in a wide number of languages in addition to English, such as Portuguese, Catalan, Romanian, Russian, French, Arabic, German, and Hebrew.¹⁰

Following their proposal, all nouns are understood to be endowed with INDEX and CONCORD features, a notion borrowed from HPSG approaches to morpho-syntactic agreement. INDEX and CONCORD features may be valued or unvalued. The innovative part of this model lies in its compositional approach to feature sharing and valuation: nouns and determiners are proposed to enter the syntactic derivation endowed with two sets of phi-features: *c*-features and *i*-features. C-features are concord features, which encode morphosyntactic properties with relevance for the phonological form (PF) interface, and *i*-features encode semantic properties inherent in lexical items with relevance for the logical form (LF) interface.¹¹ C-features include the formal, grammatical features Gender, Number, and Case, while the *i*-features include semantic properties, with features for Gender, Number, and Person. In the typical instance of nouns, such as *niña* 'girl', the feature array in (14) is proposed:

- (14) *niña* C[G(f) N(sg) C()]
 i[G(f) N(sg) P(3)]

Nouns like *mayoría* 'majority' are different in that they have the option of appearing with their *i*-features valued, similar to *niña* in (14), or unvalued, as in (15).

- (15) *mayoría* C[G(f) N(sg) C()]
 i[G(f) N(sg) P(3)] / *i*[G() N() P()]

When *i*-features are unvalued, genitive-selecting nouns like *mayoría* get their *i*-features valued by the DP-complement within the genitive KP. If that (KP-internal) NP happens to be plural, those plural number feature values become the *i*-feature values of *mayoría* as well, resulting in the ability to agree with a plural verb form. Let us consider an example of how this works with (16).

- (16) *la mayoría de los triunfadores*
 DEF.F.SG majority.F.SG of DEF.M.PL winners.M.PL
 'the majority of (the) winners'

In (16), the N head *mayoría*, which begins with the c-features [G(f), N(sg), C()] and the as yet unvalued *i*-feature set *i*[G(), N(), P()]. This syntactic object then merges with the FP *de los triunfadores* with the c-features [c[G(m) N(pl) C()] and the *i*-features [G(m) N(pl) P(3)]. Upon Agreeing, the *i*-feature values of *mayoría* become those of *los triunfadores* (15').

- (15') *mayoría* c[G(f) N(sg) C()]
i[G(m) N(pl) P(3)]

The suggestion then is that plural *i*-features for number (N) participate in agreement, at least in cases of plural verb agreement. Now, let us consider what happens when a

syntactically singular, but notionally plural DP appears within the KP, like *la mayoría del congreso* 'most of the conference'.

- (17) *congreso* c[G(m) N(sg) C()]
i[G(m) N(sg) P(3)]

When (17) merges with *mayoría*, its *i*-feature number value will become N(sg). This state of affairs should produce variable agreement for DPs like *la mayoría de las mujeres*, but never for DPs like *la mayoría del congreso*. However, a cursory Google search reveals that this prediction does not bear out, as the examples in (18) attest:¹²

- (18) a. *De todas formas no me gustan para nada los debates, y la mayoría de la gente opinan igual que yo, solo sirven para reprocharse unos a otros cuatro tonter[i]as, y los temas principales ni tocarlos, es lo mismo que pasó en el último debate Rajoy-Rubalcaba.*

'All the same, I don't like these debates at all, and **the (f.sg) majority (f.sg) of the people (f.sg) think (pl.)** the same as I do, that they only serve to berate each other with four idiotic comments, and the main issues, forget it, it's the same that happened in the last Rajoy-Rubalcaba debate'

[<https://www.foro-ciudad.com/jaen/mancha-real/mensaje-9768105.html>, Mar. 12, 2012]

- b. Pero *la mayoría de la gente no tienen* alternativas de ingreso.

'But **the (f.sg) majority (f.sg) of the (f.sg) people (f.sg) don't have (pl)**
alternative income.' [https://www.elspectador.com/economia/economia-y-
crisis-propuestas-para-enfrentar-la-pandemia-article-912345/, Apr. 1, 2020]

Therefore, it is unclear if the feature matrix in (17) for *congreso* is quite right, since either singular or plural agreement is available, similar to *gente*. If so, then it should have two possible *i*-feature sets (17'), similar to *mayoría* in (15, 15').

(17') *congreso* c[G(m) N(sg) C()]
 $i[G(m) N(sg) P(3)] / i[G() N() P()]$

In short, if singular collective nouns are also capable of valuing the *i*-feature number as plural, we expect to find similar behavior regardless of the grammatical number (c-number in this proposal) of the nouns involved, as long as semantic *i*-number may be valued as plural. As we will see in §4, such a prediction does not bear out. An additional complication is that LF processes do not typically have an influence on agreement.¹³

Pérez-Jiménez & Demonte also propose that, if no structurally lower modifier appears with plural feature values with which to value $i[...N()...]$, plural number can be anaphorically recovered. Otherwise, we would have no clear explanation for examples like (19) from Brucart (1997: 161, exs. 4a, b).¹⁴

(19) a. **La mayoría es pariente.*

DEF.F.SG majority.F.SG be.PRS.3.SG relative.M.SG

'The majority is (a) relative.'

b. *La mayoría son parientes.*

DEF.F.SG majority.F.SG be.PRS.3.PL relative.M.PL

'The majority are relatives.'

While this account fares well for examples with accompanying plural antecedents, it is unclear how semantically-plural Ns without antecedent constructions, such as *el congreso* 'conference' or *la gente* 'people', should ever have their N *i*-feature valued as plural. This state of affairs would predict uniformly singular agreement for the grammatically singular DPs in *la mayoría de la gente* 'the majority/most of the people'.

Landau (2016) also adopts the HPSG feature composition analysis of Wechsler & Zlatić (2003), by which number may be encoded as either a CONCORD feature or as an INDEX feature. In his proposal, the same distribution applies: INDEX features are semantic in nature, while CONCORD features are morphological. In the normal case, these sets of features agree, but with certain nouns, mismatches result (see Ch. 3 of Wechsler & Zlatić for numerous examples). Although the analysis in Landau focuses on particular noun-adjective mismatches in Hebrew that are largely unrelated to the current proposal, his proposal is configurational in nature in that he proposes a connection between the type of feature encoded (INDEX v. CONCORD) and its location within the DP, and it is innovative in that it seeks to account for number in distinct locations within the DP architecture. By Landau's proposal, the locus of INDEX number is NumP and the

locus of CONCORD number is NP, and each of these shares its values with INDEX and CONCORD number values on D. Given that the primary focus of his proposal is an account of variable adjective agreement properties and the sources of the underlying values, we will discuss this particular proposal no further. What is important, however, is that the notion of multiple encoding of number via semantic (INDEX) and morpho-syntactic features (CONCORD) has been previously used to explain variable agreement phenomena within the NP/DP structure. In the account that we propose in §3, we also assume that number may be encoded in different areas of the NP/DP. We also assume that number may be semantic or morpho-syntactic in nature.

3. The current proposal

Thus far, we have seen that the nouns of interest in *ad sensum* constructions are grammatically singular, yet semantically plural. Additionally, when these constructions appear as subjects, they may be accompanied by variable singular or plural verb agreement. As we saw in §2, some researchers have proposed that Number as a phi-feature may have a notional, semantic variant as well as a formal morphological variant, and these may be encoded in separate areas of the DP structure. It seems that gender and person behave similarly in this respect. In a cognitive grammar examination of agreement phenomena, Kibrik (2019: 74-75) correctly notes that, in Spanish (and Polish), third person deferential forms (i.e. *usted*, *ustedes*) are examples of *conventional* (i.e. grammatical) rather than notional agreement. Similar examples, in which a grammatically third-person pronoun is used for second-person deferential reference, are abundant in a

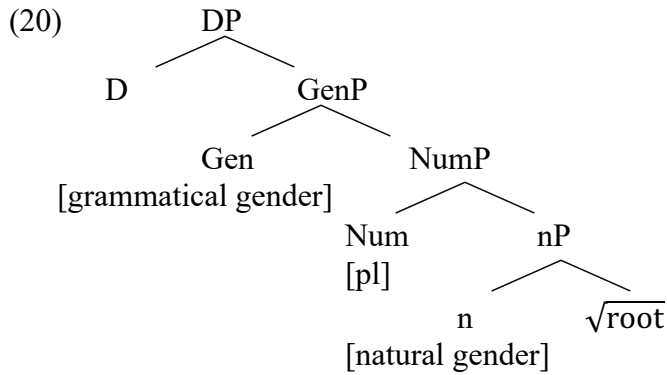
number of European languages. Kibrik also notes that gender in Russian is largely notional (i.e. semantic) with human referents.

Lowenstamm (2007) assumes a Distributed Morphology framework (e.g. Halle & Marantz 1993, Harley & Noyer 1999, Embick 2015) for building the DP (e.g. Abney 1987), arguing that the locus of gender within the DP is a low nP projection adjacent to the root. He proposes the following hierarchy of projections (20) within the DP:

$$(20) \quad DP > NumP > nP > IP > FP > nP > \sqrt{\text{root}}$$

Within this structure, FP is the locus of possessive encoding, IP is the locus of diminutive encoding, and the higher nP is the locus of features that may change a Noun's class, such as neuter gender, and NumP, which encodes plural marking.¹⁵

Kramer (2016) offers a critical evaluation of analyses that propose multiple marking of gender features within the DP. A feature of this sort of proposal is that arbitrary, grammatical gender appears low in the structure, while natural, biological gender is proposed to appear higher in the functional structure of the DP. Kramer arrives at the conclusion that hybrid agreement may merit the presence of additional, higher features in the syntax, and more research is necessary. Puškar (2018) employs a lexical decomposition approach of the type used in Distributed Morphology, proposing that natural gender is encoded in the nP projection that merges with the lexical root, as in (20), from Puškar (2018: 291, ex. 18).



We follow these proposals in assuming that Spanish nouns, which are marked for number as well as gender, have a similar structure, but augmented with natural, semantic number also being encoded in nP, while grammatical number is encoded in NumP.

Caha (2022) proposes a nanosyntactic hierarchy of projections (21) in order to account for different noun types and their associated syntactic projections within the DP structure where NumP is located.

(21) PLP > CLP > MASSP > NP

Caha proposes that syntactic Number should be decomposed into the three projections related to plural (PLP), count (CLP), and mass (MASSP) nouns. For Puškar, PLP is the locus of bare plural encoding and CLP is the locus of count class or classifier morphology encoding. Caha, however, proposes that CLP is the position of nouns that appear following numerals.¹⁶ Given that the next highest number-related projection (PLP) hosts bare plurals, which may not appear with numerals, there is a separate projection for numerals called NumP, hierarchically sandwiched between PLP and CLP. Given that

numerals will be merged in NumP, we re-locate the locus of grammatical number valuation to CLP. MASSP is the locus of mass encoding of mass nouns like *agua* ‘water’. We follow Distributed Morphology (DM) assumptions regarding merge such that roots are successively merged, moving leftward up the tree as features are valued by functional projections present within the DP spinal structure. However, following Caha, we reason that these associated positions only appear when present in a given derivation. As such, in the derivation of a count noun, MASSP will not be present. The following is the hierarchy of projections that we assume in our analysis:

$$(22) \quad DP > \text{GenP} > \text{PLP} > \text{NumP} > \text{CLP} > \text{MASSP} > \text{nP} > \sqrt{\text{root}}$$

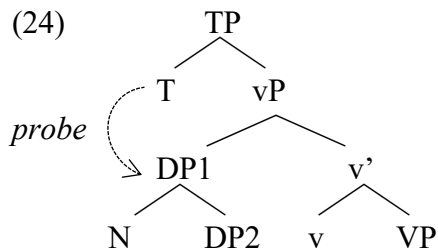
In the case of nouns, a given root will merge with an n categorial head, which is valued for semantic number and gender features. We assume that number may be valued as plural [pl] or lack a value, which triggers the singular default form; gender may be valued as feminine [f] or lack a value, which triggers the masculine by default. Given that the research surveyed above has argued for separate projections dedicated to encoding semantic as well as morphological/grammatical number and gender, we include both features in our proposal. Given that we are hopeful that the mechanism that we propose may be of use for predicting variable gender agreement in gender as well, we provide gender values as well as an invitation to readers to test the explanatory possibilities of our proposal with other variable agreement phenomena, which space considerations preclude.

We have assumed thus far that number and gender may be encoded in distinct projections. Therefore, we assume that nouns are endowed with two sets of number and gender features: semantic number (sNum), semantic gender (sGen), morphological number (mNum), and morphological gender (mGen). The highest DP-internal functional projection subsequently merges with D. For Harley & Noyer (1999: 4) closed-class Vocabulary Items like Ds express “purely grammatical properties” or “meanings determined solely by universal cognitive categories”. We follow Schoorlemmer (2012) in assuming that definiteness [def] may be encoded in D. We interpret “purely grammatical properties” as meaning that D may receive one value for number (i.e. Num) and one value for gender (i.e. Gen) upon merge with the highest maximal projection within the functional structure of the noun phrase. In our proposal, the fact that only one Num (and Gen) value may be provided to D provides the impetus for the syntax to examine its features and recognize a mismatch. Instead of the derivation crashing, a tally of the features is taken. Were D able to be valued for mNum and sNum, it is unclear to us what would motivate the system to adjudicate on how to value D. If both features can be present, it is unclear why production percentages would ever differ from 50%. However, a problem would remain: the presence of superfluous features would pose a fatal problem for the spell out of D. Following DM assumptions, the bundle of features in D references corresponding Vocabulary Items, and these exponents are inserted into the structure. We propose that the vocabulary insertion rules for determiner system for definite and indefinite articles in Spanish consists of Vocabulary Items with the following feature compositions and exponents:

(23)	<u>exponent</u>	<u>features</u>
	/un/	↔
	/una/	↔ [f]
	/unos/	↔ [pl]
	/unas/	↔ [f], [pl]
	/el/	↔ [def]
	/la/	↔ [def], [f]
	/los/	↔ [def], [pl]
	/las/	↔ [def], [f], [pl]

It is unclear how late insertion would deal with conflicting features within D, but if plural semantic number were able to be valued in addition to singular morphological number, it would seem to predict the (over)generation of impossible and infrequent sequences consisting of a female plural determiner */las/* merged with a collective singular noun, as in **/las gente/*.

Assuming a version of the clausal syntax that weds Baker's (1988) UTAH with a predicate-internal subject syntax (e.g. Koopman & Sportiche 1991), an agentive DP subject is base generated in Spec, vP, and this DP is subsequently probed by T after T merges with vP (24).



Viewed within the Agree-based framework of Chomsky (1998), agreement is a function of c-command. At this point in the derivation, DP1 will satisfy T's strong D-feature and DP1 will provide a value for morphological agreement processes on the finite verb. In Spanish, the relevant morphological feature for subject-verb agreement is number (though note that gender is relevant for subjects of passive constructions). Now, recall that *ad sensum* structures are variable in their singular or plural verb agreement. Following the standard assumption that D is valued for morphological number by its mutually c-commanding functional projection, D should only receive a value for agreement that is identical to that of its complement. Once D's number is determined, this is the value that DP provides for T. In short, if D is singular, T will end up being singular, and the same will apply for plural agreement. So, how does variable agreement ever happen? Why does it seem like the number of DP2 in (24) plays a role in agreement with T? We propose that, when D merges with its complement, D is able to see that its complement is endowed with conflicting semantic and morphological number. We also propose that this conflict triggers an operation that we name *Count* (25).

(25) *Count*

When feature valuation encounters conflicting semantic and morphological features, search for and count relevant features within the structure within its c-command domain to generate a relative frequency ratio with variable agreement instructions.

The *Count* operation recognizes the importance of the conflicting semantic information and makes a note of it in the form of a diacritic, which is a proportional calculation based on the sum of relevant features, both semantic and morphological, within the c-command domain of the workspace/derivation. This diacritic provides important semantic information for the interface with the late insertion process for functional morphemes, like *Ds*, as assumed by DM approaches (e.g. Halle & Marantz 1993, Harley & Noyer 1999, Embick 2015). This information provides guidelines for acceptable rates of variable agreement based on the featural makeup of the subject DP. In the remainder of this section, we provide sample derivations for a variety of possible subject NP/DP structures, focusing on the application of the *Count* mechanism and the Late Insertion diacritic that it provides. Each of these diacritics will provide a predicted percentage of variable agreement, which we will test in §4. We start with two examples from above (12a, b): the partitive structure (26a=12a) and the pseudo-partitive structure (26b=12b).

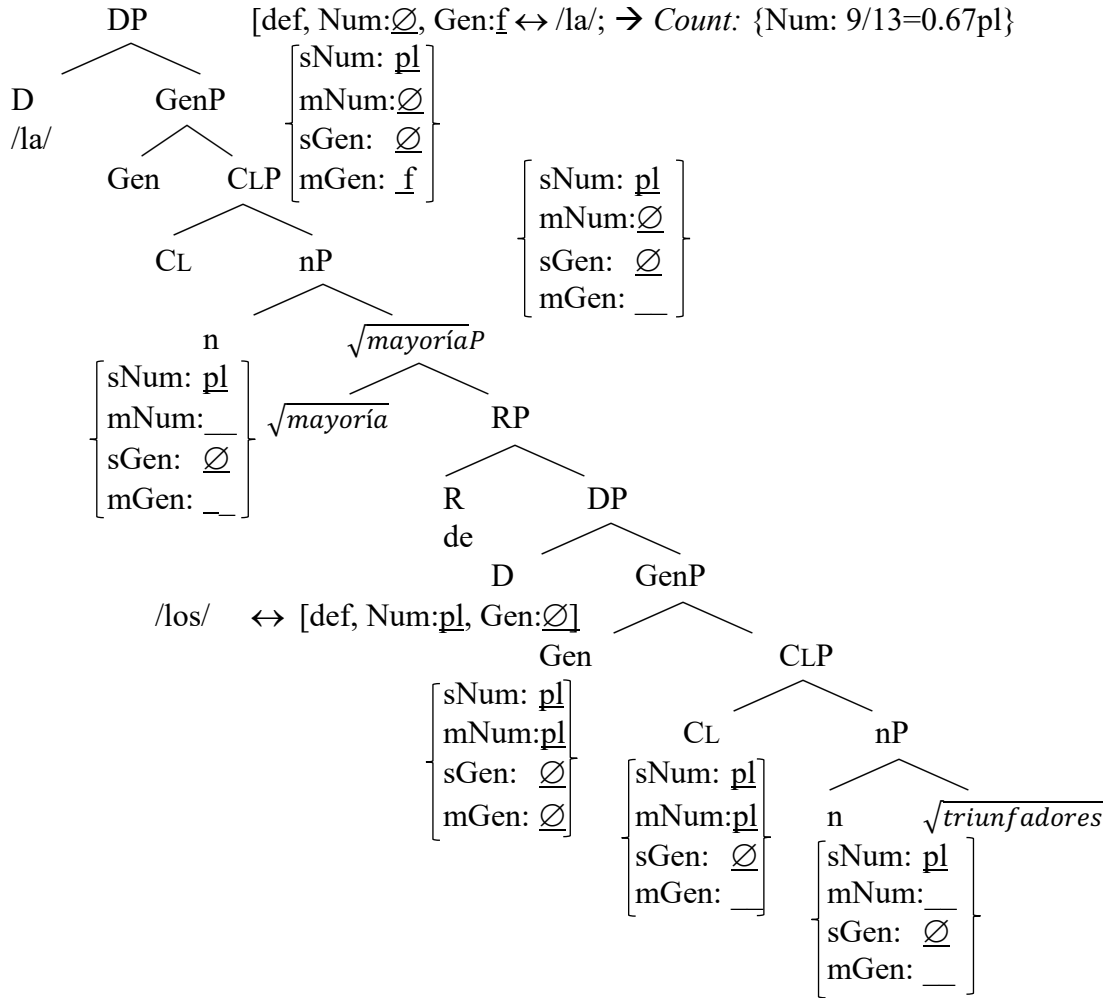
- (26) a. *La mayoría de los triunfadores*
DEF.F.SG majority.F.SG of DEF.M.PL winners.M.PL
'The majority of the winners'
- b. *Un puñado de soldados*
DEF.M.SG handful.M.SG of soldiers.M.PL
'A handful of soldiers'

Since we are using an underspecification approach to the featural makeup of definite and indefinite determiners (23), an unvalued feature for number (Num) receives the default value of singular.¹⁷ An unvalued feature for gender (Gen) bears a value of masculine, given its status as the default in Spanish (e.g. Bull 1965, Harris 1991). In the following derivations, we show features with a blank underline (e.g. mNum:) to show that a feature has not yet had the opportunity to be valued at that point of the derivation.¹⁸

In the derivation of (27), the root $\sqrt{\text{triunfadores}}$ ‘winners’ merges with the n head, which is endowed with potential values for gender and number. Following the conclusions of proposals reviewed in §2, we assume that semantic number and gender are valued in this initial merge with n as part of the semantic makeup of the nominalizer. In the next operation, the root left-adjoins to n⁰, and this structure left-adjoins to CLP for valuation of morphological number as (mNum: pl). Crucially for this proposal, the nP that left-adjoins to CLP is a copy of nP, following the tenets of the Copy Theory of Movement (Chomsky 1993, see also Nunes 2011). We propose that each subsequent copy leaves behind a record of its formal feature array, and this record is what, in our proposal, informs our understanding of defaults via underspecification as well as the syntactic building blocks of variable agreement phenomena. The CLP structure subsequently left-adjoins to GenP for valuation of masculine morphological gender as (mGen:). GenP then merges with D, which bears a definiteness feature [def]. Once gender and number are valued on D, we have the building blocks for the late insertion exponent /los/. The DP then merges with the relator (R) *de*, as proposed in Espinal & Cyrino (2022: 192-193). The relator treats the structurally higher N (i.e. within DP1) as a sort of quantifier present

in partitive constructions, which as the reader will recall, are characterized by the presence of a determiner within DP2. From here, the Relator Projection (RP) merges with the higher DP1 root $\sqrt{\text{mayoría}}$, which subsequently merges with the nominalizer head n^0 , bearing semantic features for number (sNum:pl) and gender. As $\sqrt{\text{mayoría}}$ does not bear biological gender, it receives a default value (sGen:∅). nP then merges with CLP, valuing morphological number as the unmarked default (mNum:∅). The structure then merges with Gen, valuing morphological gender (mGen:f). Now, at this point in the derivation, GenP merges with D and should value Num and Gen. However, the problem is that GenP bears conflicting sNum and mNum specifications. Although the D will be valued with the mNum value on GenP (here, mNum:∅), we propose that Count applies at this point, tallying up the total number of plural features within the c-command domain of D as a record of the plural semantics of the nominal subject. The total of plural features is nine out of a total of thirteen, a number that is converted to a decimal ratio {Num: 0.67pl}, which serves as instructions for variation in verb agreement in production. In order to test the explanatory value of Count, we will use this value as the predicted percentage of plural agreement in the corpus in §4. Prior to doing this however, we present a number of additional structures under examination in the current paper in order to clarify how Count is implemented and what structures display opacity to Count.

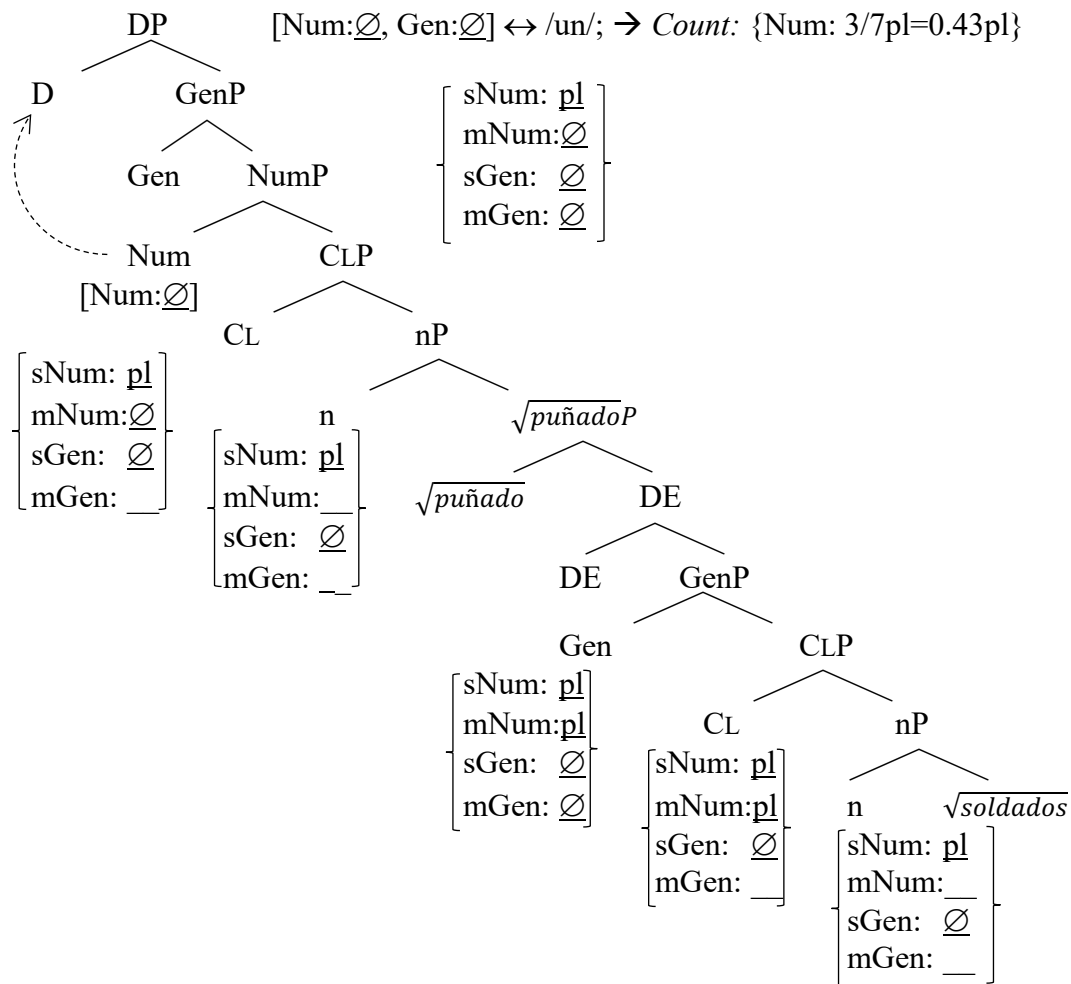
(27) *La mayoría de los triunfadores.*



We now examine a pseudo-partitive structure in (28). The merge procedure advances similarly to that of (27). However, the structurally-lower NP2 *de soldados* in GenP merges with a *DE* operator (Espinal & Cyrino 2022: 193-203) that cancels definiteness. This operator then merges with $\sqrt{\text{puñado}}$, which afterwards merges with its nominalizer particle. The n^0 is valued as plural for semantic number (sNum:pl) and remains unmarked for semantic gender (sGen: $\varnothing$). Merge with CL^0 values morphological number with a default. In the next merge step, CLP merges with the numeral head, valuing it for default

morphological number (Num: $\emptyset$). The syntactic object NumP then merges with Gen, and mGen is valued (mGen: $\emptyset$). At this point, the numeral moves to D, but given the conflicting semantic and morphological number values in GenP, Count is once again invoked. We claim that it is this operator that makes the lower NP2 opaque to Count, thus tallying three plural features out of a possible seven within the c-command domain of D, generating a predicted frequency proportion of {Num: 0.43pl}.

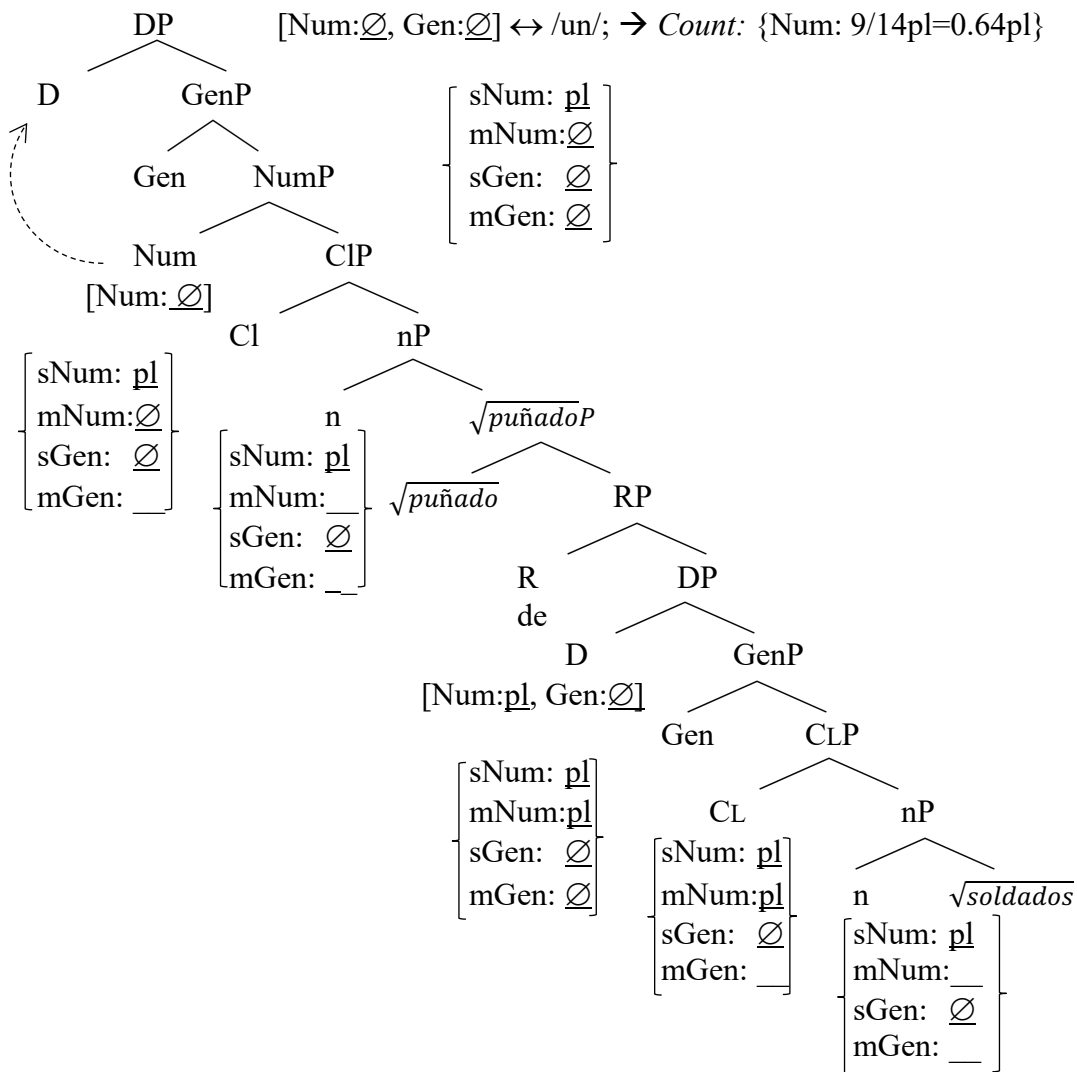
(28) *Un puñado de soldados*



If the DE operator were not opaque to Count, we would expect this calculus to produce a different production frequency ratio that is more similar to the partitive structure in (27).

Consider (29):

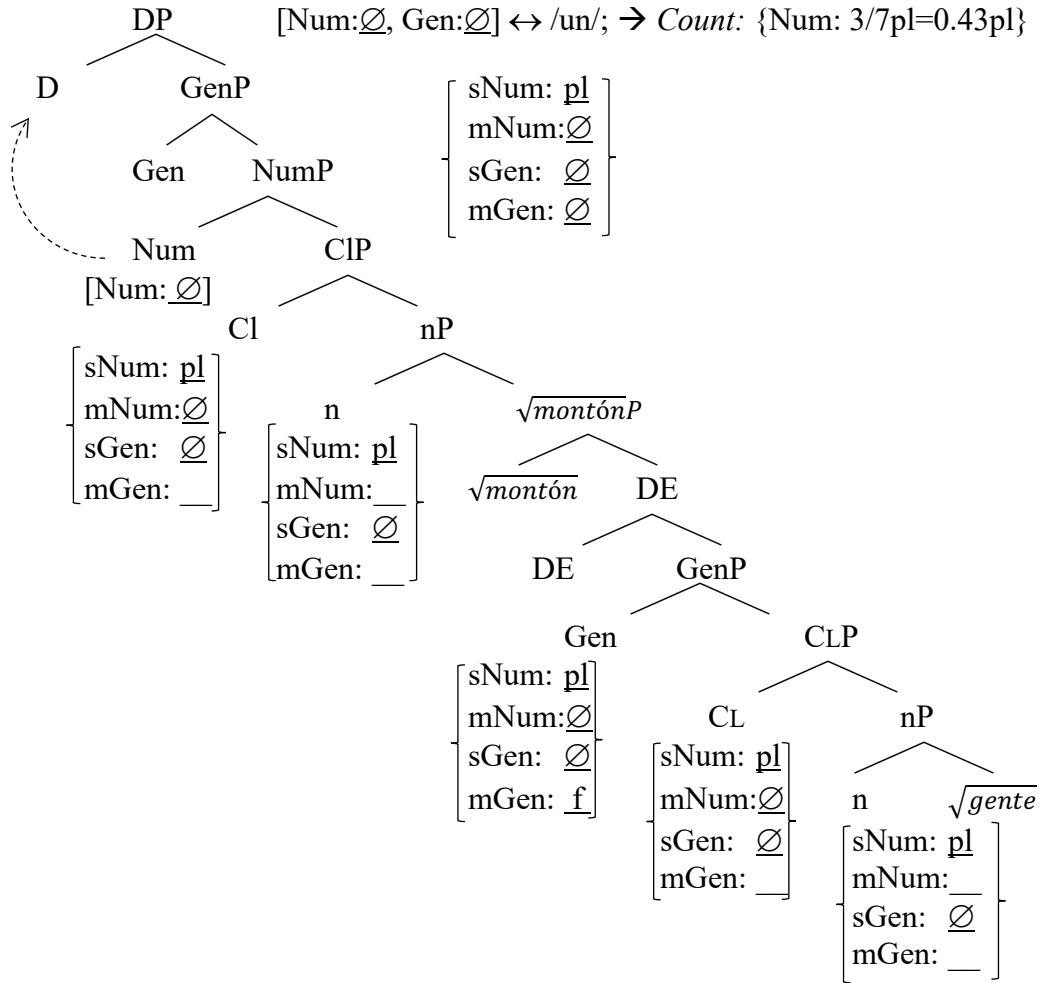
(29) *Un puñado de los soldados*



Now, if (28) has a partitive structure, like (29), we predict that Count will tally a total of eight plural features out of a possible thirteen (Num=0.62pl), excluding the lone plural number feature in the D^0 of DP2 in (29). As we will see in §4, the corpus results favor an approach by which pseudo-partitives are opaque to Count. This receives a straightforward explanation with the adoption of the DE operator.

In (30), we examine a pseudo-partitive structure with singular nouns with plural semantics. As in (29), the features relevant for Count upon merge of GenP with D are only those within the highest DP (i.e. DP1). In this particular structure, Count identifies three plural features out of a possible seven valued for $\sqrt{montón}$, thus generating a frequency ratio of {Num=0.43pl}, similar to (28).

(30) *Un montón de gente*

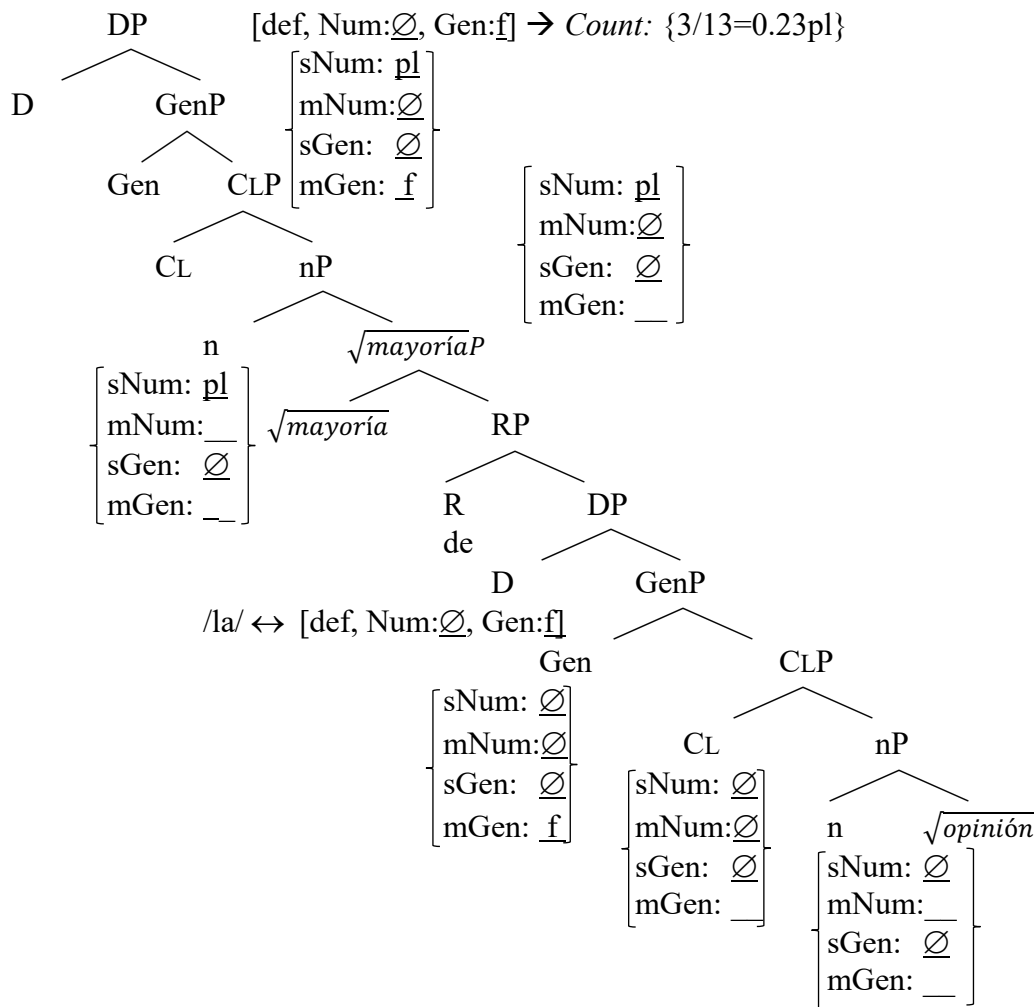


The final sample derivation we offer prior to presenting the empirical test of our proposal using corpus data is of a partitive structure with collective, semantically-plural nouns in order to illustrate how these features are calculated.

In the final partitive structure that we examine, we can see the influence that semantic and morphological features are predicted to have on variable agreement. Here, we start with the root $\sqrt{\text{opinión}}$, which merges with the n^0 head, valuing semantic

number (sNum:∅) and semantic gender as (sGen:∅), reflecting the fact that *opinión* ‘opinion’ is not semantically plural or collective, and bears no semantic, biological gender specification.

(31) *La mayoría de la opinión*¹⁹

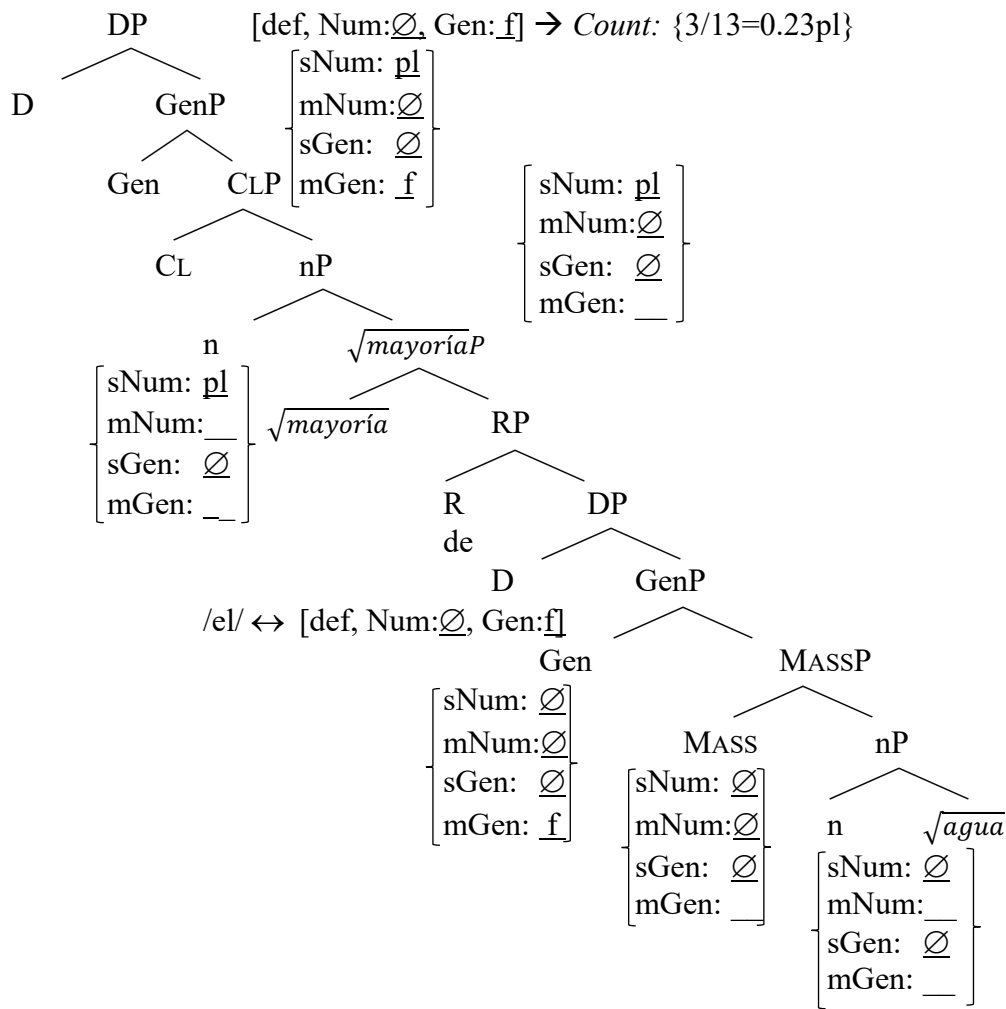


nP subsequently merges with CL⁰, valuing morphological number (mNum:∅), and CLP then merges with Gen, valuing morphological gender (mGen:f). Upon merge with D, with bears [def], Num is valued (Num:∅) and gender is valued (Gen:f), a combination

that will lead to the exponent /la/ at late insertion. This merges with Espinal & Cyrino's (2022) relator *de*, projecting RP, which then merges with the structurally higher root $\sqrt{\text{mayoría}}$. Upon merger with the nominalizer (n^0), semantic features for number and gender are valued as (sNum:pl) and (sGen: $\emptyset$), respectively. The structure then merges with CL^0 , valuing morphological number (mNum: $\emptyset$), and then with Gen^0 , valuing morphological gender (mGen: $\underline{f}$). When the highest GenP merges with D, a conflict results between possible values for number, triggering Count. Count subsequently tallies three plural markings out of a possible thirteen, resulting in a frequency ratio of {Num=0.23pl}.

For mass nouns in expressions like *la mayoría del agua* 'the majority of the water', we propose a slightly different derivation. As discussed above, we follow Caha (2022) in assuming that the locus of mass nouns in the syntax is MASSP. In this derivation, nP left-adjoints to MASS, thus encoding *agua* as a mass noun.²⁰ This expression never appears with plural verb agreement. In this case, we suspect that when Count proceeds to tally number features from within the c-command domain of D, it finds the MASSP projection within DP2. Within this structure, it finds (adjoined) nP. When no featural conflict between mNum and sNum is present, it halts and produces a frequency ratio for plural of '0'.

(32) *La mayoría del agua*



To summarize, in this section, we have laid out our proposal for how Count works and have provided sample derivations and the production ratios generated by Count that should be reflected in statistically similar ratios in production. In the following section, we detail our methodology for testing the predictions of our proposal with data from the *Corpus del Español* (Davies 2016).

4. Testing the predictions of the model with corpora

The *Count* operation that we have proposed works in the spirit of Adger's (2006) *combinatorial variability* to "predict (rather than capture) frequencies of variants".²¹ Adger (2006) examines variable subject-verb agreement in Buckie Scottish English (e.g. *we were* vs. *we was*) adopting a fairly standard Minimalist feature-checking system as in Chomsky (1995). Importantly, this variation is not associated with a variable meaning, nor is it influenced by social factors. In his account, lexical items may be underspecified for uninterpretable phi-features, and this is what can generate patterns of variability and categoricity, or invariability. He proposes that the following minimal phi-features are associated with a verb-form exponent (33):

- (33) a. [*usingular*: +] was
 b. [*usingular*: -] were
 c. [*uparticipant*: +] was
 d. [*uauthor*: -] were
 e. [*uauthor*: +] was

For the subject pronoun *you* [plural], the phi-feature set (34) results and, as we can see, this particular phi-combination has two paths to generating *was* from (33).

- (34) [*singular*: +, *participant*: +, *author*: -] (a) was, (c) was, (d) were

According to his proposal, the relevant feature matrices predict frequencies of variation as they are expected to occur in spoken corpora gathered in a typical sociolinguistic interview. In this case, two out of three outcomes result in *was*, so the predicted frequency should be 67%. In the corpus examined, the attested frequency of second-person singular *was* was 69%, which is almost identical to the predicted outcome. Although the sample size was small, combinatorial variability is a potentially promising heuristic for capturing and predicting agreement variation.

In order to test the predictions of our proposal, we accessed the 2-billion-word web portion of the *Corpus del Español* (Davies 2016, henceforth CDE) using the Corpus Query Processor (Hardie, 2012, henceforth CQP), which allows for corpus queries using part-of-speech tags in addition to specific lexical items. The POS tags used for the nouns in this corpus contain morphological information (i.e. number and gender) that was leveraged in our queries. We sought to perform searches for two main DP1 *de* NP2/DP2 structures in Spanish that participate in *ad sensum* phenomena:

- 1) DP_{indef} + *de* + NP/DP (e.g. *una mayoría de*+NP/DP ‘a majority of+NP/DP’)
- 2) DP_{def} + *de* + NP/DP (e.g. *la mayoría de*+NP/DP ‘the majority of+NP/DP’)

We conducted searches for singular definite determiners (*el, la*) and indefinite determiners (*un, una*) within DP1, always using the determiner that agreed with the head noun in grammatical gender. To narrow down the scope of Nouns within DP1, we first consulted the *Nueva Gramática de la Real Academia Española* (RAE, 2009), taking inventory of all of the nouns cited as participating in partitive and pseudo-partitive constructions. We then searched for all of these nouns within the corpus. We then chose

the top 15 most frequent nouns for targeted corpus queries, with eight nouns classified as participating in partitive structures (see Table 1) according to the RAE, and seven nouns classified as pseudo-partitive (see Table 2), netting a total of 7,712,167 occurrences in the corpus.

Spanish noun	rank	freq total
<i>parte</i> ‘part’	96	1548121
<i>millón</i> ‘million’	208	774213
<i>mayoría</i> ‘majority’	275	571447
<i>número</i> ‘number’	278	562097
<i>cantidad</i> ‘quantity’	361	446607
<i>resto</i> ‘rest’	546	311358
<i>total</i> ‘total’	747	233008
<i>par</i> ‘pair’	809	215179
total		4,662,030

Table 1. Eight most frequent nouns associated with partitivity in the CDE by rank and frequency totals.

Spanish noun	rank	freq total
<i>grupo</i> ‘group’	166	930527
<i>familia</i> ‘family’	190	805663
<i>comunidad</i> ‘community’	313	498116
<i>serie</i> ‘series’	397	412432
<i>conjunto</i> ‘ensemble’	792	219528
<i>ejército</i> ‘army’	1620	103783
<i>montón</i> ‘heap’	2012	80088
total		3,050,137

Table 2. Seven most frequent nouns associated with pseudo-partitivity in the CDE by rank and frequency totals.

We then conducted twelve searches for each partitive and pseudo-partitive head noun.

Each consisted of the following linear structures summarized in Table 3.

DP1		de	DP2
<i>el/la</i>	(Noun)	<i>de</i>	sg. bare NP
			pl. bare NP
			def. masc. sg. DP
			def. fem sg. DP
			def. masc. pl. DP
def. fem. pl. DP			
<i>un/una</i>			sg. bare NP
			pl. bare NP
			def. masc. sg. DP
			def. fem sg. DP
	def. masc. pl. DP		
			def. fem. pl. DP

Table 3. Possible DP1 *de* DP2 combinations and their predicted agreement frequencies, according to the proposed operation *Count*.

Given that the corpus is not tagged for hierarchical structure, we were forced to search for the linear sequence: (Det)+NP1+*de*+(Det)+NP2+V.²² Once we tallied the data from the searches, we had to make some adjustments, owing to the coding of the corpus. All NPs had to be searched separately, by noun gender, but gender is not relevant for the subject-verb agreement process in Spanish. Therefore, we manually combined similar construction totals, effectively eliminating gender as a variable for singular DP2s as well as plural DP2s for each DP1 determiner type, thus leaving us with 8 possible combinations per (pseudo)partitive noun type, in addition to the predicted percentages of verb agreement following *Count*. Now, as we have proposed in §3, if the lower NP2 structure in pseudo-partitives is opaque to *Count*, we expect different rates of plural expression in partitive structures as compared to pseudo-partitive structures. The possible

combinations and corresponding predicted *Count* proportions are summarized in Table 4, with the example head nouns *montón* ‘heap’ and *mayoría* ‘majority’.²³

DP1 de NP2/DP2		predicted verb agreement values	
		% sg.	% pl
<i>el /la</i>	sg. bare NP2	50	50
	pl. bare NP2	50	50
	def. sg. DP2	77	23
	def. pl. DP2	33	67
<i>un / una</i>	sg. bare NP2	57	43
	pl. bare NP2	57	43
	def. sg. DP2	57	43
	def. pl. DP2	36	64

Table 4. Simplified possible DP1 *de* NP2/DP2 combinations and their predicted agreement frequencies, according to the proposed operation *Count*.

4.1. Descriptive results

We have created contingency tables in order to report all of the possible combinations based on the results of the corpus searches. Table 5 shows the numbers for the most frequent nouns appearing in DP1 *de* NP2/DP2 constructions that we extracted from the corpus, while Table 6 shows the numbers for the most frequent nouns in pseudo-partitive DP1 *de* NP2/DP2 constructions that we extracted from the corpus. In the interest of transparency, we report on all possible DP1 *de* NP2/DP2 structures extracted from the corpus. In other words, we do not just report on DP1 *de* NP2 structures for nouns associated with pseudopartitive structures.

First, we notice that there are over three times as many partitive noun constructions as there are pseudo-partitive constructions. Additionally, we can see that partitive structures

occur more often with definite determiners than pseudo-partitive structures do. We can also see that certain DP1 nouns, like *mayoría* ‘majority’, *grupo* ‘group’, and *resto* ‘rest’ have over one thousand occurrences, while others, like *total* ‘total’, or *ejército* ‘army’ are much less represented, despite their overall frequency. An additional striking observation is that certain combinations of (pseudo)partitive noun type and NP2/DP2 are more frequent than others, and not all possible combinations examined here are equitably represented in Tables 5 and 6.

DP1	finite verb agreement (3 rd pers.)	el/la DP1 de NP2sg	el/la DP1 de NP2pl	el/la DP1 de DP2sg	el/la DP1 de DP2pl	un/una DP1 de NP2sg	un/una DP1 de NP2pl	un/una DP1 de DP2sg	un/una DP1 de DP2pl	total
parte	sg	57	20	88	24	7	0	483	45	724
	pl	9	6	19	4	0	2	36	35	111
millón	sg	0	17	0	0	0	171	0	0	188
	pl	1	14	0	0	1	385	0	0	401
mayoría	sg	188	780	2343	3053	0	23	15	6	6408
	pl	30	1467	63	7341	0	22	0	8	8931
número	sg	95	1756	23	29	13	20	1	0	1937
	pl	12	94	2	6	2	38	0	0	154
cantidad	sg	279	437	19	4	17	4	0	0	760
	pl	26	67	2	0	3	10	0	0	108
resto	sg	32	288	1901	349	1	0	2	0	2573
	pl	9	559	472	536	0	0	0	0	1576
total	sg	9	97	85	28	0	0	1	0	220
	pl	0	26	11	29	1	0	0	0	67
par	sg	2	40	1	0	15	1173	1	0	1232
	pl	0	21	0	0	1	680	0	1	703
	total sg	662	3435	4460	3487	53	1391	503	51	14042
	total pl	87	2254	569	7916	8	1137	36	44	12051
	TOTAL	749	5689	5029	11403	61	2528	539	95	26093

Table 5. Contingency table for totals of singular and plural verbs with partitive nouns in eight *DP1 de NP2/DP2* structures in the CDE.

Notably, we can see in the detailed summaries by partitive noun construction in Table 5 that 7,341 instances of plural verb agreement involve *mayoría* followed by a plural DP2, while 1,467 instances of plural agreement follow a plural NP2.²⁴ In total, constructions with *mayoría* constitute over 15,000 of the total 26,093 occurrences of these structures. With respect to singular verb agreement, 2,343 cases involve *mayoría* modified by a singular DP2, while 1,901 instances involve *resto* modified by a singular DP2. In Table 6 we can see that among pseudo-partitive nouns, *grupo* followed by a plural NP2 results in 1,390 cases of plural agreement and 2,308 instances of singular agreement.

DP1	finite verb agreement (3 rd pers.)	el/la DP1 de NP2sg	el/la DP1 de NP2pl	el/la DP1 de DP2sg	el/la DP1 de DP2pl	un/una DP1 de NP2sg	un/una DP1 de NP2pl	un/una DP1 de DP2sg	un/una DP1 de DP2pl	total
grupo	sg	265	352	31	18	183	2308	22	5	3184
	pl	90	82	7	17	22	1390	1	3	1612
familia	sg	19	10	217	23	5	20	4	0	298
	pl	5	8	35	15	2	0	0	0	65
comunidad	sg	52	108	34	19	7	37	0	0	257
	pl	17	20	23	5	6	6	0	0	77
serie	sg	19	34	5	4	21	265	1	1	350
	pl	3	9	1	1	7	391	0	0	412
conjunto	sg	24	87	154	47	2	78	1	0	393
	pl	2	32	18	15	0	72	0	0	139
ejército	sg	13	9	21	12	2	16	1	1	75
	pl	2	2	3	0	0	7	0	0	14
montón	sg	1	2	0	0	70	47	2	0	122
	pl	0	3	0	0	5	112	1	1	122
total sg		393	602	462	123	290	2771	31	7	4679

total pl	119	156	87	53	42	1978	2	4	2441
TOTAL	512	758	549	176	332	4749	33	11	7120

Table 6. Contingency table for totals of singular and plural verbs with pseudo-partitive nouns in eight *DP1 de NP2/DP2* structures in the CDE.

Given the large quantity of data and numerous variables at play in the descriptive data presented in Tables 5 and 6, we leveraged two different post-hoc statistical procedures to test the explanatory power of Count: 1) a linear regression analysis to determine if the variables predicted to be of importance by the *Count* mechanism are significant in the corpus results, and 2) a Chi-squared analysis comparing predicted outcomes based on Count to the attested corpus outcomes.

4.2. Discussion

Taken together, the corpus results numbering 33,213 total observations in tables 5 and 6 leave the state of things unclear with respect to what precise variables trigger *ad sensum* agreement, but our proposed *Count* mechanism provides us with a working hypothesis. To test our hypothesis, we ran a linear regression analysis to determine if these factors within the *DP1 de NP2/DP2* structures extracted from the corpus predict the appearance of a plural verb form.

All relevant dependent and independent variables in these observations were coded as binary values, a procedure that makes possible a linear regression analysis to determine which variable(s) are most strongly associated with the plural verb agreement indicative of *ad sensum* structures. The dependent variable was verb agreement (plural=1, singular=0). Within the *DP1 de NP2/DP2* structure, however, we want to know what

independent variables are most strongly associated with plural verb agreement in the corpus. We chose semantic features as well as features relevant to Count for our coding procedure in order to test the explanatory value of our hypothesis. The noun within DP1 was coded binarily as either partitive or pseudo-partitive (partitive=1, pseudo-partitive=0) following the classification in Tables 1 and 2. Recall from our preceding discussion, that our corpus searches were only performed on *DP1 de NP2/DP2* structures in the corpus containing the 15 most frequent nouns cited by the RAE as appearing in partitive and pseudo-partitive structures. Within DP1, all D heads were either definite or indefinite (definite=1, indefinite=0). The NP1 within DP1 was uniformly singular (despite its proposed semantic plurality) and constitutes a trivial variable, so we coded the head Noun (NHead) with a category value, in this instance, the name of the N itself. Within the NP2/DP2 modifier, the determiner could be singular, plural, or bare. However, if our proposed *Count* mechanism is on the right track in its calculus, bare NP2s will not be endowed with features that are accessible to Count for plural agreement. In other words, only the semantic and morphological features in a (non-bare) DP2 will be relevant for plural agreement. We therefore coded bare as (bare=1, non-bare=0). Finally, the NP2, independent of whether it is located within a DP2, could be morphologically singular or plural (plural=1, singular=0).

We ran a linear regression model with our codified data in R using *plm* (Croissant & Millo, 2008), using the code in (35):

(35) `plm(formula = Vpl ~ partitive + defhead + bare + Npl, data = data, model = "random", index = c("HeadN"))`

As can be seen in the R code, we also entered the head noun as a random variable. The regression results in Table 1 of Appendix 1 show that bare nouns within the NP2 ($p < 0.001$) and plural nouns within the DP2 ($p < 0.001$) are significantly associated with plural verb agreement, while definiteness approaches significance ($p = 0.096$). To determine if there was a particular association between bareness and plurality among nouns, we re-ran the model, as reported in Table 2 in Appendix 1, removing the non-significant variables *partitive* and *defhead*, but coding for an interaction between the independent variables bare nouns and plural nouns (*bare * Npl*). Regression results suggest that there is a statistically significant association between plural verbal agreement and a bare plural noun within NP2 ($p < 0.001$). Recall that, as we saw in Table 4, there are very different rates of predicted plural verb values for plural NP2s and plural DP2s. If our Count proposal is on the right track, plural NP2s should be associated with fewer instances of plural agreement owing to a differential *Count* algorithm such that NP2 is opaque to *Count*, which means that expected values are based solely on the result of *Count* within DP1. Therefore, in order to offer some clarity with respect to these variables and their predictive power as proposed in Count, we compare our predicted values with our observed values, which appear in Tables 3-10 of Appendix 1. We carried out Chi-squared comparisons of the predicted and observed values, and found that all of the results were significant ($p < 0.01$), meaning that there is no statistical difference between the expected and observed values. A graphic summary of the observed corpus verb agreement rates by nominal structure appears in Figure 1.

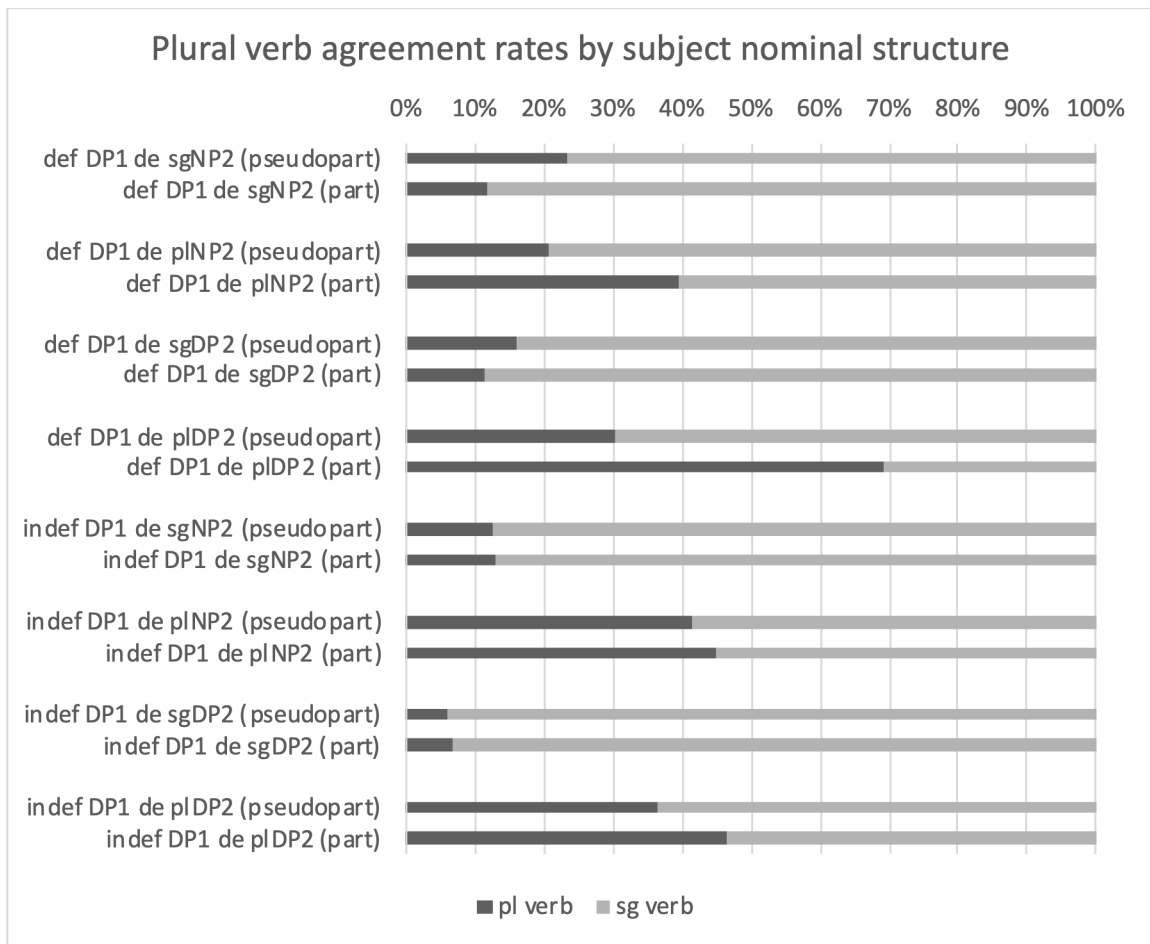


Figure 1. Plural verb agreement rates by subject nominal structure.

As an additional robustness check on the results predicted by *Count*, we ran additional corpus queries, focusing on verb agreement for a subset of the independent variables (plural DP2), but modified by an adjective agreeing in number (i.e., plural), such as *la mayoría de los estudiantes extranjeros* ‘the majority of the foreign students.’ As compared to the definite DP1 structure in (27), for example, this contributes an additional plural number feature to the *Count* mechanism, thus making the predicted

plural verb agreement rates $\{10/14=0.71\text{pl}\}$ for the definite structure and $\{10/15=0.667\text{pl}\}$ for the indefinite structure. We summarize the updated predicted rates based on the feature composition relevant to *Count* as well as the corpus search results in Table 7.

	predicted		observed		total
	singular	plural	singular	plural	
definite	447.3 (28.6%)	1116.7 (71.4%)	361 (23.1%)	1203 (76.9%)	1564
indefinite	10 (33.3%)	20 (66.7%)	19 (63.3%)	11 (36.7%)	30

Table 7. Predicted and observed verb agreement totals and percentages for definite DP1 *de* plural DP2 combinations with plural adjectives by determiner type

The total attested results in Table 7 represent higher rates over DP2 structures without an agreeing adjective, in line with the predictions of *Count* as we have proposed. A Chi-squared test confirms the statistical significance of the attested values as compared to the predicted values using the diacritic proposed to be the output of *Count* ($p < 0.001$). This result is strongly suggestive that different feature combinations like the ones examined here, with *ad sensum* structures that are augmented by features from an agreeing plural adjective within DP2, result in higher rates of plural verb agreement. The rates that we have hypothesized are confirmed by statistical analysis of the results in the dataset. In accordance with our proposal, by which the computational system is supplemented with a *Count* mechanism that provides important information to the interface regarding certain features, additional constituents with plural features influence higher rates of plural verb agreement than in structures with fewer plural features.

5. Conclusions

There is always a tension present between descriptive accuracy and explanatory adequacy in formal syntactic analysis. When we are confronted with empirical data that suggest necessary modifications for the working theoretical model, we must either modify the model to accommodate the data, or arrive at an informed decision as to why we should disregard the data and the insight that they might otherwise offer regarding modifications to the model. We feel that it is time that a generative syntactic model should be proposed that can make strides toward accounting for variable agreement data in *ad sensum* structures in languages like Spanish, thus capturing descriptive generalizations while making accurate predictions about the structure in question. We see the current study, which leverages feature matrices and their derivational copies to make such predictions, as a positive step in this direction.

The current theoretical proposal, taking advantage of the proposed *Count* operation to explain agreement variation, seeks to account for data that do not receive a satisfying theoretical explanation to date—at least with respect to frequencies in production. Here, we have attempted to motivate a feature-based mechanism, compatible with current Minimalist theorizing, called *Count* that can calculate variable agreement patterns based on a ratio of singular and plural figures within the complex structure *DP1 de DP2*. When the highest functional projection—GenP in many of the sample derivations provided in section 3—merges with D, it is tasked with valuing number features on D. The clash has two results: the D is valued with the morphological mNum

value to avoid crash, but *Count* is invoked in order to generate a variable production ratio that will be of relevance for the DP upon merging with finite T. The raw numbers in evidence in Tables 5 and 6 exhibit uneven frequencies and distributions by noun and construction type. The regression results examined helped to shed light on the situation, and confirm that the factors predicted to be significant by *Count* are operative in the data. As further confirmation, we have offered Chi-squared comparisons of predicted values generated by *Count* and observed corpus frequencies of these structures, and all test confirm the predictions of *Count*.

An obvious limitation of the current proposal has to do with the scope of languages examined. Methodologically, we chose the *CdE* as an initial test of our proposal because it is widely cited and used for corpus research on Spanish, but an additional benefit was its convenience. Unfortunately, an automated process capable of verifying part of speech and hierarchical structure does not currently exist for this corpus because it is not tagged for such information. A corpus that was tagged for syntactic hierarchy, part of speech, and partitivity would prove much more precise. The examination of a number of noun types, such as animate nouns and nouns like *clase/tipo* ‘class/type’, and how these factors might affect their behaviors, may prove to be productive. For the moment, corpora that are not tagged for such features must be searched manually. Many of these nouns appeared in our early searches of the RAE included such nouns. However, we restricted our corpus searches to the most frequent nouns in our lists to avoid any appearance of ‘cherrypicking’, or choosing nouns within the DP1 that would be conducive to some particular, predetermined outcome. Given

current technological limitations, we mined the RAE (2009) for the most common nouns that participate in partitive and pseudo-partitive structures, leading to the perhaps unfortunate association of nouns with structures in Tables 1 and 2. Notwithstanding, a further examination of the difference in N type and their frequency and association with different structures was curious, and merits further investigation. Note that, for definite DP1 *de* plural DP2 (reported in Appendix 1, Table 6), a partitive construction, the vast majority of the results were for head nouns that were reported to be associated with partitive constructions: in this case, 11403/11579 definite DP1 *de* DP2 partitive structures had partitive nouns in the DP1. For certain constructions, it seemed that the noun type favored structure type, but for others, the trend is opposite. However, certain other construction types were rather infrequent. Nevertheless, the overall numbers of partitive and pseudo-partitive constructions examined in our corpus search were very similar (Table 8, see Appendix 1 for full report of results).

structure	corpus total
pseudopartitive (DP1 <i>de</i> NP2)	15378
partitive (DP1 <i>de</i> DP2)	17835
TOTAL	33213
DP1 determiner type	corpus total
indefinite	8348
definite	24865
TOTAL	33213

Table 8. Corpus totals by structure type and DP1 determiner type.

Future investigations should examine differences between definite and indefinite DP1 structures in greater depth, especially given that indefinites were used much less frequently overall (Table 7, 8). It may also prove fruitful to consider the possibility of geographical variation among varieties of Spanish, especially given well-known differences such as the differential behavior of *committee* in British versus American English (see e.g. Gardelle 2019). Future studies should also replicate this study with similar *ad sensum* structures in languages endowed with them, taking into account the relevant features proposed to reside in the clausal subject. Count may also prove to be productive in explaining variable gender agreement phenomena in Slavic languages, or other languages found to exhibit the 3/4 agreement pattern referenced in a number of studies (e.g. Smith 2015, 2021; Landau 2016). Such tests will go a long way in confirming with greater precision the theoretical motivation and empirical necessity for a *Count* mechanism. If we find similar evidence in crosslinguistic data, we can replicate our search on similar corpora. Similar results would also provide us with important insight on the nature of features and the roles that they play in the syntax and whether *ad sensum* phenomena are subject to cross-linguistic variation. One might be tempted to ask if a change favoring singular agreement is part of an organic historical change, or if normative, prescriptive forces play a role in accelerating this sort of outcome by societal stigmatization of *ad sensum* agreement patterns.²⁵ We are currently unable to address such questions, but leave a deeper examination of language variation and change related to this structure for future research. Charles Yang's variational approach to language and

associated Tolerance Principle (e.g. Yang 2002, 2016) may prove critical in arming predictions for acquisition of these structures and their historical evolution.

There was a time when experimental testing of generative theory was uncommon, and it may eventually be the case that corpus data is also used to test as well as inform syntactic theory. Certainly, limitations will arise with respect to testing infrequent structures. Leal Méndez & Slabakova (2019), for example, found that clitic left-dislocation structures and contrastive-corrective contrast constructions were extremely infrequent in a Spanish corpus. As we have seen, *ad sensum* structures in general are not terribly infrequent, but certain types of them are more frequent than others. We have not proposed a syntactic model of these structures that is driven by frequency of the type that Guy (2015) would seem to favor; rather, we have proposed one that is able to generate usage guidelines for individual variation in usage based on number features that permeate the structure, a notion which may prove similar to what Newmeyer (2015: 42) had in mind when he suggested that “a grammatical system interacting with usage-based facts, but not itself incorporating those facts, is the best-motivated approach.”

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¹ We have translated the Spanish *montón* as ‘heap’ to more accurately capture its conversational equivalent in English.

² See Corbett (2000), Gardelle (2019) for further discussion.

³ Within the psycholinguistic literature, there is a similar phenomenon, often referred to as *attraction*. It has been examined in a number of languages including English (e.g. Bock & Miller 1991), French (e.g. Franck, Frauenfelder & Rizzi 2007; Franck & Wagers 2020), German (e.g. Hartsuiker, Schriefers, Bock & Kikstra 2003), Russian (e.g. Slioussar 2018), Slovenian (e.g. Marušić, Nevins & Badecker 2015), Serbian (e.g. Ristić, Molinaro & Mancini 2016), and Spanish (Fuchs, Polinsky & Scontras 2015; Lago, Chalom, Sigman, Lau & Phillips 2015). For Phillips & Lewis (2013: 23), apparent attraction effects are “grammatical illusions”. It is worth noting, however, that the term *attraction* is problematic in that it does not refer to a uniform phenomenon, and may be understood in multiple ways. While a more in-depth review of this literature is beyond the scope of this paper, in this study we limit our attention to grammatical instances of *ad sensum* agreement, and do not examine ungrammatical structures like (i).

(i) *The key to the cabinets are hard to reach.

⁴ We do not review Selkirk (1977) here primarily for reasons of space, especially given that the X’-framework in that proposal is anachronistic at present. We refer the interested reader to the source material.

⁵ See Selkirk (1977) for additional discussion related to partitives and pseudo-partitives in English.

⁶ See also Jackendoff (1968) for similar structures and definiteness effects in English.

⁷ Examples of the *ad sensum* phenomenon in Spanish and its multiple subject-verb agreement possibilities are also reported in Gili Gaya (1980: 31) in the following example (i):

- (i) *La mitad de los náufragos se salvaron /*
 DEF.F.SG half.F.SG of DEF.M.PL castaways.M.PL SE save.PST.3.PL
se salvó.
 SE save.PST.3.SG
 'Half of the castaways were saved.'

⁸ Here we use MP (Measure Phrase) and FP (Functional Phrase) as short-hand, stand-in categories to illustrate the locality problem. We discuss the precise structure of the DP that we adopt in section 3.

⁹ Owing to space concerns, we restrict our review of Demonte & Pérez-Jiménez (2015, 2017) to the theoretical proposal. We refer the interested reader to their presentation and discussion of corpus data, which differs in numerous ways from our corpus approach.

¹⁰ Slioussar (2018) in particular cautions that attraction phenomena in Russian differ from *ad sensum* phenomena in other languages. See also fn. 2.

¹¹ For more on this proposed feature matrix, couched within an HPSG framework, see Wechsler & Zlatic (2000, 2003), Wechsler (2011).

¹² These Google search results are from i) a message board about politics from Spain (20a) and ii) from a Colombian newspaper (20b). Here, in order to streamline the presentation, we combine glosses and idiomatic translation, highlighting the relevant agreement phenomena in bold.

¹³ We thank **READER COLLEAGUE** for bringing this to our attention.

¹⁴ An open empirical question is what sort of variation is present among predicates. Example (i) (Brucart's (3b)) suggests that the predicate must play some role in this process.

- (i) *La mayoría consiguió/ consiguieron el triunfo.*
 the majority achieve.PST.3.SG / achieve.PST.3.PL the triumph
 'Most were able to triumph'

This state of affairs may mean a slightly different model of S-V agreement for copula, which may shed light on the puzzling grammaticality asymmetry in (19). We leave this issue for future research. The agreement asymmetry may be related to what Coon & Keine (2021) call hierarchy effects of person and number features with copula in examples, similar to (ii-iv) in Spanish.

- (ii) *El inteligente *es/soy yo.*
 the intelligent be.PRS.3.sg / be.PRS.1.sg I
 'I am the smart one'
 (iii) *Los malos *son/sois vosotros*
 the bad be.PRS.2pl / be.PRS.3pl you.all
 'You all are the bad guys'
 (iv) *El problema *es/son ellos*
 the problem be.PRS.3.sg / be.PRS.3.pl they
 'The problem is them'

Following their proposal, the most specified of these features should prevail in agreement terms within copula. We are grateful to **READER COLLEAGUE** for this observation and the examples.

¹⁵ See also Ritter (1991), Bernstein (1993), Picallo (2008) for previous approaches to functional projections within the DP field. The higher nP here appears to differ from the proposal in Satik (2017), where the higher nP appears to the left of DP and is necessary for capturing languages with multiple determiners, such as Brazilian Portuguese (BP), which allows sequences like *o/este meu cachorro* 'the/this my dog'.

¹⁶ This may be the same projection as Measure Phrase (MP) in Stickney (2009).

¹⁷ We depart from Fuchs et al (2015: 709) on this point. They claim that the plural is semantically unspecified. We treat the singular as the default option because it is the least-marked form morphologically among nouns and verbs in Spanish. See also fn. 16. See Noyer (1997) for an underspecification account applied to Ugaritic prefixes.

¹⁸ As we will see, morphological features are unvalued on initial merge and remain so for several steps in the derivation. We believe that this is a productive way in which to see the pervasive influence of default forms within the morpho-syntax.

¹⁹ Certain structural configurations, in particular ‘*mayoría* + DP2’ structures, are not possible in Spanish. This example comes directly from the *CdE* corpus.

²⁰ See e.g. Corbett (2000), Soames (2002) for further discussion of the semantics of mass nouns.

²¹ See also Smith & Adger (2006), Adger & Smith (2010).

²² We are aware that structures like ‘*el montón de* def.sg.DP’ are arguably not partitive sequences. As a result of the fact that corpora are not currently tagged for partitivity, we are currently obliged to carry out corpus searches in this way, paying attention exclusively to linear order of lexical material. For this reason, in order to steer our searches toward structures that were more likely to be (pseudo)partitives, we searched for the head nouns attested in RAE (2009). In the interest of transparency, we provide examples of each structure recovered from the corpus in Appendix 2.

²³ One may wonder what makes *mayoría* ‘majority’ semantically plural. According to a selection of semantic accounts, such as Gardelle (2019: 74-75), *majority* in English is a prototypical collective noun. For Levin (2001: 49), *majority* in English often appears with plural verbs. The same seems to apply for Spanish. See both for further discussion of these issues. Although collective nouns may be ambiguous between a collective and distributive reading, an exploration of this possibility is orthogonal to the current manuscript, so we leave it for further research. For an interesting proposal on collective nouns like *equipo* ‘team’ as partitive constructions that may be understood as either individuals or as a collective whole, see Areses & Pérez-Martínez (2022). If our proposal is on the right track, these nouns should also trigger Count since they also bear conflicting values of morphological and semantic number. We leave a more in-depth exploration of how to incorporate this proposal for future research.

²⁴ Although we use the linear notion *follow*, we do not suggest that linear adjacency is necessarily indicative of hierarchical syntactic adjacency.

²⁵ We refer readers interested in the diachronic syntax of partitives and pseudo-partitives to Rutkowski (2007), who works within the formal grammaticalization assumptions of Roberts & Roussou (1999).